2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.202 Human-Computer Interaction

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

1.a In the context of the design thinking process, explain:

i. The purpose of the 'define' step.

[1 mark]

ii. Why the 'define' step is important.

[2 marks]

iii. What happens during the 'define' step.

[2 marks]

1.b Describe two need-finding tools that allow you to observe longitudinal or sporadic behaviour.

[4 marks]

1.c. You wish to compare the performance of two different physical keyboard layouts, QWERTY and alphabetical.

i. Briefly explain the suitability of performing a Keystroke-Level Model (KLM) analysis to determine the performance differences between the two keyboards.

[4 marks]

ii. You decide to run a controlled experiment where 20 participants type the same 50 sentences using each keyboard. Name the two most useful pieces of quantitative data that you will collect during the study to help you determine the performance differences between the two keyboards.

[2 marks]

iii. Describe one characteristic of the 20 participants that will heavily influence the outcome of the study.

[1 mark]

iv. To complement the experimental data, and to help you make an overall recommendation, you also wish to gather self-reported feedback on the participants' experience. Describe two qualitative data points you would collect and a suitable method for gathering this information.

[4 marks]

Total 20 marks

Question 2

2a. Why should we study Human-Computer Interaction? Frame your answer around the three views, Economic, System, and Social, providing two reasons for each viewpoint.

[6 marks]

2b. Touchless interfaces:

i. Describe two categories of interaction supported by touchless interfaces, and for each, provide an example that already exists.

[2 marks]

ii. For each category, describe two challenges those interfaces face during real-world implementation.

[4 marks]

2c. Fitts' Law can be used to predict the movement time for a well-rehearsed selection task using the formula MT = a + b x ID.

i. Describe the relationship between distance and target size.

[2 marks]

ii. You wish to use Fitts' Law to compare the effectiveness of pointing with a new device, SuperMouse, with that achieved using a traditional mouse. Describe an experimental setup that would allow you to collect the necessary data for a comparison based on Fitts' Law.

[4 marks]

iii. From your study you observe that the traditional mouse adheres to Fitts' law, whereas SuperMouse does not. What can you conclude from this result?

[2 marks]

Total 20 marks

Question 3

3a. Provide BRIEF answers to the following questions.

i. What is the economic principle of prototyping, in the simplest terms?

[1 mark]

ii. What do "elaboration" and "reduction" refer to in the context of design processes?

[2 marks]

iii. Prototypes are a means for communication. Who are the audiences?

[2 marks]

3b. Consider the application you designed in your coursework, or any other application of interest. Provide a high-level description of the application in 1-2 sentences.

i. Sketch a horizontal prototype of the application: describe what the prototype would look like, and what it would convey.

[2 marks]

ii. Sketch and describe a vertical prototype of the same application. Explain how it would differ from the horizontal prototype, and elaborate the respective advantages of horizontal and vertical prototypes.

[3 marks]

3c. In your coursework, you used paper and click prototyping in the process of designing an interactive application.

i. Describe what paper prototyping is good for and provide a concrete example from your coursework experience that highlights a benefit of the technique.

[3 marks]

ii. Consider the limitations of paper prototyping and give one concrete example of a design question you were not able to explore with the technique.

[3 marks]

iii. Given the design question you described, what other prototyping technique could you have used to explore it? Justify your choice by explaining what the technique is good for, in general, and describe how you could have used it in your coursework project.

[4 marks]

Total 20 marks

Question 4

4a. Provide BRIEF answers to the following questions:

i. Identify the three processors in the MHP model of human information processing.

[1.5 marks]

ii. Identify Gestalt principles that describe what we perceive to be organised in groups.

[1.5 marks]

iii. What is the difference between mistakes and slips?

[2 marks]

4b. A general design principle is to reduce the demands of an interface on the user's short term memory.

i. Explain what short-term memory is.

[3 marks]

ii. Psychology research in the 1950s led to the theory that people can hold up to seven plus minus two items of information in their working memory. Many designers have since adopted the "magical number seven plus minus two" as a limit for the number of options presented in a menu, and the number of bullets in list. Explain why this is an inappropriate application of the theory.

[2 marks]

iii. The previous question refers to the limited capacity of short term memory. Identify and explain at least one other characteristic of short-term memory that has implications for user interface design.

[2 marks]

iv. What are mode errors, how are they linked to short-term memory, and how can they be avoided?

[3 marks]

4c. Responsiveness of a system is perhaps the most important factor for user satisfaction. Provide a definition for responsiveness and discuss how it relates to but is different from performance of a system. Give an example of a design guideline that relates to responsiveness.

[5 marks]

Total 20 marks.

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